Supplementary Information

Benchmarking the accuracy, robustness and reproducibility of spatial domain identification methods

Supplementary Notes

Supplementary Note 1. Data and preprocessing

DLPFC count matrices from spatialLIBD and layer annotations from the SEDR analyses repository; STARmap and MERFISH matrices and annotations from the BASS analyses. Breast cancer: 10x Genomics Space Ranger 1.1.0 dataset V1_Breast_Cancer_Block_A_Section_1 with the 20-region annotation (column fine_annot_type) of the SEDR analyses repository (data/BRCA1/metadata.tsv). Stereo-seq: MOSTA atlas files E9.5_E1S1, E10.5_E2S1 and E10.5_E1S1 (bin50, CNGB STOmics DB STDS0000058), raw counts and the published tissue annotation (obs annotation); bins without counts were removed. The two later datasets were added after the analysis of the first three was complete and run with unchanged settings in the standardised runner (three seeds per method; two for CCST on E10.5_E1S1). Genes in fewer than 50 spots or with fewer than 10 counts were removed for the whole-transcriptome data (Visium, Stereo-seq); 2000 highly variable genes were selected on the counts (Seurat v3 flavour) or all panel genes kept (STARmap, MERFISH), then counts were normalised to the library size, log-transformed, scaled and reduced by PCA (200 components for whole-transcriptome data, 50 for panels). Sections, splits and label provenance are listed in configs/datasets.yaml; per-section statistics are in Supplementary Table S1.

Supplementary Note 2. Methods and settings

Every method was run from its official code with the settings of its tutorial or paper (Supplementary Table S2). The true number of domains was given to every method. Neighbourhood radii given in pixels were converted to multiples of the spot or cell spacing so that they transfer across technologies. Methods that output an embedding were additionally clustered with matched clustering: a 20-start tied-covariance Gaussian mixture keeping the best likelihood, the same for every method.

Supplementary Note 3. Metrics

Agreement: ARI, NMI, AMI. Per-domain: clusters matched to domains with the Hungarian algorithm; macro-F1, missed domains, recall of domains below 5% of spots. Spatial: CHAOS (mean distance to the nearest same-cluster neighbour), PAS (fraction of spots whose cluster differs from most of their ten neighbours), Moran's I of the labels, boundary F1. Label-free: silhouette, Davies–Bouldin, Calinski–Harabasz. Definitions and code: spatialbench/metrics.py (unit tests in tests/).

Supplementary Note 4. Statistics

Sections are nested in groups (three DLPFC donors, STARmap, MERFISH, breast cancer, Stereo-seq). Differences to the reference are equal-weight means of group means with 95% percentile intervals from a hierarchical bootstrap (5000 resamples: groups, then sections within groups). Equivalence: 90% interval within ± 0.03 ARI, about the seed-to-seed standard deviation. No correction for multiple comparisons is applied. Ranks and the probability of ranking first come from the same resamples. The reference method was chosen after seeing the means, so these comparisons are exploratory. Code: `scripts/stats_cluster_aware.py`.

Supplementary Note 5. Audit of released code

Repositories were screened with `scripts/audit_leakage.py` for five patterns of label use (Supplementary Table S20), and every candidate line was read by one author. The taxonomy and all judgements, including false alarms, are in `docs/LEAKAGE_AUDIT.md`. The audit describes what the released code does; it does not establish how published results were obtained.

Supplementary Note 6. Computing environment

All runs used one CPU thread per job on a 4-core Intel Xeon (2.8 GHz) with 15 GB of memory, Python 3.13, R 4.3.3 and mclust 6.0.1. Memory-heavy methods (DeepST, BayesSpace, NichePCA) were run one job at a time. For the breast cancer and Stereo-seq runs every job ran alone with a cap of 12 GB resident memory (process tree, sampled once per second); a job above the cap, or leaving less than 0.8 GB of system memory, was stopped and recorded as out of memory.

Supplementary Tables

Supplementary Table S1. Data and quality control per section. Sparsity: fraction of zero entries. Annotated: fraction of spots or cells with a domain label. Smallest domain: fraction of annotated spots in the smallest domain. Imbalance: largest over smallest domain size.

Section	Spots/cells	Genes	Median counts	Sparsity	Annotated	Domains	Smallest domain	Imbalance
DLPFC 151507	4226	33538	2464	0.96	1.00	7	0.072	4.0
DLPFC 151508	4384	33538	2019	0.96	1.00	7	0.046	6.9
DLPFC 151509	4789	33538	2416	0.96	1.00	7	0.035	11.3
DLPFC 151510	4634	33538	2319	0.96	0.99	7	0.039	9.9
DLPFC 151669	3661	33538	3470	0.95	0.99	5	0.063	9.3
DLPFC 151670	3498	33538	3113	0.95	1.00	5	0.060	10.4
DLPFC 151671	4110	33538	3743	0.94	1.00	5	0.060	7.8
DLPFC 151672	4015	33538	3482	0.95	0.97	5	0.078	5.2
DLPFC 151673	3639	33538	4120	0.93	0.99	7	0.060	4.5
DLPFC 151674	3673	33538	5334	0.92	0.99	7	0.062	4.1
DLPFC 151675	3592	33538	3140	0.95	0.99	7	0.077	2.8
DLPFC 151676	3460	33538	3419	0.94	0.99	7	0.074	3.3
STARmap 20180417_BZ5_control	1049	166	604	0.69	1.00	4	0.082	5.2
STARmap 20180419_BZ9_control	1053	166	301	0.77	1.00	4	0.078	6.1
STARmap 20180424_BZ14_control	1088	166	393	0.70	1.00	4	0.031	13.6
MERFISH −0.04	5488	155	241	0.58	1.00	8	0.019	13.1
MERFISH −0.09	5557	155	260	0.57	1.00	8	0.025	9.7
MERFISH −0.14	5926	155	246	0.57	1.00	8	0.036	7.8
MERFISH −0.19	5803	155	254	0.58	1.00	8	0.043	5.6
MERFISH −0.24	5543	155	244	0.59	1.00	8	0.041	7.0
Breast block_a_1	3798	36601	20762	0.85	1.00	20	0.007	16.0
Stereo-seq E9.5_E1S1	5913	25568	14039	0.86	1.00	12	0.017	15.5
Stereo-seq E10.5_E2S1	8494	22385	7102	0.91	1.00	18	0.013	13.1
Stereo-seq E10.5_E1S1	18408	25201	11786	0.86	1.00	13	0.011	23.5

Supplementary Table S2. Benchmarked methods: code source, version, settings source, clustering and every deviation from the published tutorial (from configs/methods.yaml).

Method	Type	Code	Commit	Settings	Default clustering	Deviations from the tutorial
HiSTaR	deep graph VAE	PyTorch port of the official code (hicast/histar.py)	0039a3b	paper	3-start Gaussian mixture (scikit-learn, tied)	masking at inference removed in 'deterministic' variant only; headline uses published behaviour; per-section lambda_sim tuning described in the paper's supplement is NOT applied
GraphST	deep graph contrastive autoencoder	pip GraphST 1.1.1; tutorials from JinmiaoChenLab/GraphST	d62b0b7	tutorial	R mclust EEE on 20 PCs of the embedding	reported with and without the tutorial's label refinement (graphst vs graphst_refined)
BANKSY	neighbourhood-augmented features (non-deep)	pip pybanksy 1.3.5	9278996	tutorial	3-start Gaussian mixture (scikit-learn, tied)	both recommended lambdas are run: 0.2 (cell typing) and 0.8 (domain segmentation)
SpaGCN	graph convolutional network	pip SpaGCN 1.2.7	dc7a1c2	tutorial	iterative DEC clustering (own)	no histology image used (not available for all sections); reported with and without the hexagonal refinement
STAGATE	deep graph attention autoencoder	QIFEIDKN/STAGATE_pyG, installed from the clone	ae1158c	tutorial	R mclust EEE	radius expressed in spot spacings because our coordinates are on another scale (gives 5.8 neighbours per spot, as in the tutorial); torch_sparse replaced by a stub only if the compiled extension is unavailable (it was compiled from source here)
SEDR	deep masked graph autoencoder	JinmiaoChenLab/SEDR, installed from the clone	ef48360	tutorial	R mclust EEE (seed 2023)	mclust called as SEDR.mclust_R does, without its line that hard-codes the authors' R_HOME (breaks R's LAPACK in a running session)
SpaceFlow	deep graph infomax with spatial regularisation	pip SpaceFlow 1.0.4	58d1ab0	package default	Leiden at a fixed resolution	networkx>=3 alias for removed to_scipy_sparse_matrix; all-zero genes removed (cell_ranger HVG binning fails on them with pandas 3)
NichePCA	training-free (neighbour aggregation + PCA)	pip nichepca 0.1.0	package 0.1.0	package default	Leiden	
CCST	deep graph infomax (4 GCN layers)	xiaoyeye/CCST, re-implemented in scripts/new_methods.py because of PyTorch 1.7 pins	890de27	tutorial	k-means (n_init=100)	graph threshold chosen to give about 6 neighbours per cell (the authors hand-pick a threshold per dataset)
Spatial-MGCN	deep multi-view graph convolution	cs-wangbo/Spatial-MGCN, repository modules used directly	cf4412d	tutorial	k-means each epoch; epoch chosen by ARI against the annotation (LEAKAGE, see docs/LEAKAGE_AUDIT.md)	unannotated spots kept (the released script deletes them before training); label-free epoch rule
DeepST	deep graph + augmentation (no histology)	JiangBioLab/DeepST (deepstkit), used from the clone	e37997f	tutorial	Leiden at the given K + spatial refinement	no histology features; memory-heavy: run one job at a time
BASS	Bayesian hierarchical model	zhengli09/BASS, installed from the clone	5c2690f	tutorial	own (joint cell-type and domain model)	each section analysed separately (the authors' MERFISH/DLPFC analyses are multi-sample)
BayesSpace	Bayesian statistical model	edward130603/BayesSpace tag v1.5.1 (older release; newer ones need unavailable dependencies)	ba8b422	package default	own MCMC	Visium only (needs a hexagonal spot lattice)
Linear diffusion	training-free			fixed rule	3-start Gaussian mixture (scikit-learn, tied)	settings fixed on DLPFC donor 1 before any other data were analysed
AnisoST	training-free, edge-preserving diffusion			fixed rule	3-start Gaussian mixture (scikit-learn, tied)	same as linear_diffusion
PCA	training-free			fixed rule	3-start Gaussian mixture (scikit-learn, tied)	

Supplementary Table S3. Mean ARI over sections and seeds per dataset with each method's default clustering and with matched clustering (the same 20-start best-likelihood Gaussian mixture on every embedding). The training-free baselines, HiSTaR and BANKSY use a 3-start Gaussian mixture as default clustering. SpaGCN, BASS and BayesSpace cluster internally and have the same value under both protocols.

Method	DLPFC default	DLPFC matched	STARmap default	STARmap matched	MERFISH default	MERFISH matched	Breast default	Breast matched	Stereo-seq default	Stereo-seq matched
Linear diffusion	0.543	0.558	0.708	0.705	0.481	0.497	0.546	0.545	0.306	0.306
AnisoST	0.526	0.551	0.564	0.622	0.377	0.363	0.568	0.544	0.357	0.346
PCA	0.388	0.409	0.270	0.294	0.151	0.130	0.555	0.562	0.348	0.349
HiSTaR	0.491	0.544	0.593	0.658	0.221	0.264	0.528	0.530	0.320	0.307
GraphST	0.514	0.531	0.388	0.484	0.251	0.254	0.497	0.577	0.346	0.315
BANKSY λ=0.8	0.494	0.518	0.740	0.694	0.428	0.410	0.610	0.605	0.320	0.298
BANKSY λ=0.2	0.514	0.518	0.502	0.529	0.211	0.222	0.590	0.582	0.341	0.341
SpaGCN	0.422	0.422	0.267	0.267	0.151	0.151	0.578	0.578	0.388	0.388
STAGATE	0.497	0.548	0.479	0.572	0.368	0.342	0.483	0.557	0.353	0.326
SEDR	0.515	0.551	0.593	0.236	0.406	0.167	0.416	0.522	0.281	0.241
SpaceFlow	0.312	0.326	0.628	0.668	0.389	0.260	0.452	0.546	0.333	0.323
NichePCA	0.321	0.506	0.692	0.604	0.475	0.403	0.538	0.614	0.305	0.358
CCST	0.437	0.359	0.505	0.435	0.212	0.157	0.525	0.564	0.284	0.271
Spatial-MGCN	0.418	0.391	0.303	0.245	0.182	0.133	0.575	0.552	0.271	0.280
DeepST	0.459	0.435	0.368	0.409	0.231	0.163	0.560	0.552	0.319	0.365
BASS	0.484	0.484	0.716	0.716	0.582	0.582	0.624	0.624	0.352	0.352
BayesSpace	0.437	0.437	–	–	–	–	0.564	0.564	–	–

Supplementary Table S4. As Supplementary Table S3 for normalised mutual information (NMI).

Method	DLPFC default	DLPFC matched	STARmap default	STARmap matched	MERFISH default	MERFISH matched	Breast default	Breast matched	Stereo-seq default	Stereo-seq matched
Linear diffusion	0.668	0.676	0.700	0.699	0.602	0.614	0.712	0.714	0.534	0.534
AnisoST	0.653	0.665	0.582	0.590	0.494	0.488	0.675	0.677	0.569	0.563
PCA	0.497	0.506	0.301	0.326	0.227	0.216	0.637	0.637	0.554	0.556
HiSTaR	0.637	0.660	0.602	0.653	0.375	0.405	0.692	0.684	0.554	0.551
GraphST	0.621	0.634	0.435	0.481	0.408	0.404	0.646	0.669	0.574	0.550
BANKSY λ=0.8	0.605	0.611	0.713	0.680	0.567	0.558	0.711	0.710	0.547	0.533
BANKSY λ=0.2	0.610	0.611	0.530	0.535	0.302	0.318	0.693	0.694	0.564	0.564
SpaGCN	0.555	0.555	0.317	0.317	0.238	0.238	0.656	0.656	0.603	0.603
STAGATE	0.641	0.668	0.492	0.552	0.490	0.473	0.694	0.706	0.573	0.554
SEDR	0.654	0.654	0.614	0.303	0.544	0.312	0.650	0.677	0.517	0.501
SpaceFlow	0.456	0.480	0.640	0.655	0.509	0.441	0.641	0.653	0.558	0.554
NichePCA	0.460	0.629	0.703	0.604	0.567	0.540	0.721	0.702	0.553	0.582
CCST	0.576	0.511	0.453	0.421	0.334	0.314	0.625	0.655	0.511	0.504
Spatial-MGCN	0.599	0.581	0.307	0.271	0.265	0.232	0.652	0.687	0.526	0.541
DeepST	0.628	0.572	0.489	0.429	0.370	0.261	0.688	0.651	0.584	0.563
BASS	0.639	0.639	0.690	0.690	0.638	0.638	0.694	0.694	0.599	0.599
BayesSpace	0.597	0.597	–	–	–	–	0.679	0.679	–	–

Supplementary Table S5. Difference in mean ARI between linear diffusion and each method, both with their default clustering; equal-weight mean of group means; 95% interval from a hierarchical bootstrap over seven groups (three DLPFC donors, STARmap, MERFISH, breast cancer, Stereo-seq); per-dataset columns give the mean difference within each dataset. Test only: the same estimate without the development sections. Verdict: baseline higher if the interval is above zero; unresolved otherwise; no comparison met the equivalence criterion (90% interval within ± 0.03).

Method	Δ	95% CI	Bonferroni CI	DLPFC	STARmap	MERFISH	Breast	Stereo-seq	Test only	95% CI	Verdict
PCA	+0.169	(+0.050, +0.296)	(+0.007, +0.364)	+0.16	+0.44	+0.33	−0.01	−0.04	+0.173	(+0.040, +0.316)	baseline higher
Spatial-MGCN	+0.152	(+0.052, +0.271)	(+0.016, +0.322)	+0.12	+0.41	+0.30	−0.03	+0.01	+0.159	(+0.043, +0.286)	baseline higher
SpaGCN	+0.146	(+0.021, +0.279)	(−0.028, +0.350)	+0.12	+0.44	+0.33	−0.03	−0.08	+0.152	(+0.008, +0.299)	baseline higher
SpaceFlow	+0.134	(+0.059, +0.217)	(+0.028, +0.255)	+0.23	+0.08	+0.09	+0.09	−0.03	+0.133	(+0.046, +0.223)	baseline higher
CCST	+0.120	(+0.054, +0.191)	(+0.027, +0.227)	+0.11	+0.20	+0.27	+0.02	+0.02	+0.105	(+0.036, +0.189)	baseline higher
DeepST	+0.114	(+0.020, +0.220)	(−0.015, +0.265)	+0.08	+0.34	+0.25	−0.01	−0.03	+0.127	(+0.013, +0.250)	baseline higher
NichePCA	+0.097	(+0.012, +0.186)	(−0.019, +0.221)	+0.22	+0.02	+0.01	+0.01	−0.02	+0.068	(−0.030, +0.175)	baseline higher
GraphST	+0.090	(−0.003, +0.193)	(−0.031, +0.259)	+0.03	+0.32	+0.23	+0.05	−0.06	+0.089	(−0.011, +0.203)	unresolved
HiSTaR	+0.077	(+0.008, +0.160)	(−0.019, +0.199)	+0.05	+0.12	+0.26	+0.02	−0.01	+0.072	(−0.016, +0.167)	baseline higher
BayesSpace	+0.075	(−0.003, +0.163)	(−0.025, +0.211)	+0.11	–	–	−0.02	–	+0.080	(−0.018, +0.186)	unresolved
STAGATE	+0.071	(+0.001, +0.145)	(−0.028, +0.184)	+0.05	+0.23	+0.11	+0.06	−0.05	+0.073	(+0.007, +0.137)	baseline higher
BANKSY λ=0.2	+0.069	(−0.011, +0.167)	(−0.034, +0.220)	+0.03	+0.21	+0.27	−0.04	−0.04	+0.064	(−0.015, +0.157)	unresolved
SEDR	+0.062	(+0.016, +0.110)	(−0.003, +0.140)	+0.03	+0.12	+0.08	+0.13	+0.03	+0.057	(+0.012, +0.104)	baseline higher
AnisoST	+0.033	(−0.016, +0.086)	(−0.033, +0.115)	+0.02	+0.14	+0.10	−0.02	−0.05	+0.032	(−0.015, +0.079)	unresolved
BANKSY λ=0.8	+0.013	(−0.030, +0.055)	(−0.048, +0.074)	+0.05	−0.03	+0.05	−0.06	−0.01	+0.007	(−0.043, +0.059)	unresolved
BASS	−0.011	(−0.067, +0.051)	(−0.088, +0.088)	+0.06	−0.01	−0.10	−0.08	−0.07	−0.022	(−0.079, +0.050)	unresolved

Supplementary Table S6. Primary analysis: as Supplementary Table S5 with matched clustering (the same 20-start best-likelihood Gaussian mixture on every embedding; SpaGCN, BASS and BayesSpace use their internal clustering).

Method	Δ	95% CI	Bonferroni CI	DLPFC	STARmap	MERFISH	Breast	Stereo-seq	Test only	95% CI	Verdict
Spatial-MGCN	+0.188	(+0.066, +0.324)	(+0.015, +0.400)	+0.17	+0.46	+0.36	−0.01	−0.00	+0.204	(+0.063, +0.353)	baseline higher
CCST	+0.175	(+0.067, +0.274)	(+0.018, +0.315)	+0.20	+0.27	+0.34	−0.02	+0.04	+0.169	(+0.049, +0.288)	baseline higher
PCA	+0.167	(+0.049, +0.289)	(−0.007, +0.353)	+0.15	+0.41	+0.37	−0.02	−0.04	+0.165	(+0.030, +0.303)	baseline higher
SpaGCN	+0.154	(+0.025, +0.289)	(−0.027, +0.346)	+0.14	+0.44	+0.35	−0.03	−0.08	+0.160	(+0.009, +0.317)	baseline higher
SpaceFlow	+0.136	(+0.041, +0.226)	(−0.003, +0.263)	+0.23	+0.04	+0.24	−0.00	−0.02	+0.132	(+0.012, +0.243)	baseline higher
SEDR	+0.130	(+0.005, +0.269)	(−0.035, +0.350)	+0.01	+0.47	+0.33	+0.02	+0.07	+0.148	(−0.002, +0.325)	baseline higher
DeepST	+0.129	(+0.022, +0.239)	(−0.019, +0.291)	+0.12	+0.30	+0.33	−0.01	−0.09	+0.119	(+0.003, +0.244)	baseline higher
BayesSpace	+0.086	(+0.002, +0.178)	(−0.021, +0.219)	+0.12	–	–	−0.02	–	+0.088	(−0.019, +0.200)	baseline higher
BANKSY λ=0.2	+0.071	(−0.011, +0.169)	(−0.042, +0.237)	+0.04	+0.18	+0.28	−0.04	−0.04	+0.063	(−0.023, +0.165)	unresolved
GraphST	+0.068	(−0.015, +0.161)	(−0.046, +0.205)	+0.03	+0.22	+0.24	−0.03	−0.04	+0.056	(−0.029, +0.154)	unresolved
HiSTaR	+0.048	(−0.012, +0.125)	(−0.036, +0.163)	+0.01	+0.05	+0.23	+0.01	−0.00	+0.047	(−0.036, +0.141)	unresolved
STAGATE	+0.041	(−0.011, +0.101)	(−0.028, +0.138)	+0.01	+0.13	+0.15	−0.01	−0.02	+0.044	(−0.008, +0.100)	unresolved
NichePCA	+0.029	(−0.035, +0.092)	(−0.062, +0.125)	+0.05	+0.10	+0.09	−0.07	−0.08	+0.019	(−0.044, +0.086)	unresolved
AnisoST	+0.029	(−0.015, +0.080)	(−0.033, +0.106)	+0.01	+0.08	+0.13	+0.00	−0.04	+0.032	(−0.014, +0.090)	unresolved
BANKSY λ=0.8	+0.024	(−0.020, +0.069)	(−0.040, +0.092)	+0.04	+0.01	+0.09	−0.06	+0.01	+0.027	(−0.025, +0.082)	unresolved
BASS	−0.004	(−0.063, +0.061)	(−0.084, +0.095)	+0.07	−0.01	−0.09	−0.08	−0.07	−0.016	(−0.077, +0.059)	unresolved

Supplementary Table S7. Sensitivity analysis: as Supplementary Table S6 (matched clustering) but with every dataset taken from the standardised runner (one execution environment, shared preprocessing), and the corresponding bootstrap ranks on the sections completed by every method.

Method	Δ	95% CI	Bonferroni CI	DLPFC	STARmap	MERFISH	Breast	Stereo-seq	Verdict
Spatial-MGCN	+0.184	(+0.060, +0.316)	(+0.019, +0.375)	+0.16	+0.46	+0.36	−0.01	−0.00	baseline higher
PCA	+0.172	(+0.053, +0.301)	(−0.004, +0.354)	+0.15	+0.44	+0.37	−0.02	−0.04	baseline higher
SpaGCN	+0.169	(+0.035, +0.319)	(−0.020, +0.383)	+0.15	+0.47	+0.38	−0.03	−0.08	baseline higher
CCST	+0.158	(+0.061, +0.251)	(+0.015, +0.295)	+0.16	+0.28	+0.33	−0.02	+0.04	baseline higher
SpaceFlow	+0.145	(+0.051, +0.235)	(+0.012, +0.277)	+0.24	+0.07	+0.24	−0.00	−0.02	baseline higher
DeepST	+0.143	(+0.026, +0.266)	(−0.018, +0.321)	+0.14	+0.35	+0.33	−0.01	−0.09	baseline higher
SEDR	+0.139	(+0.013, +0.288)	(−0.023, +0.360)	+0.02	+0.50	+0.33	+0.02	+0.07	baseline higher
BayesSpace	+0.094	(+0.005, +0.184)	(−0.019, +0.222)	+0.13	–	–	−0.02	–	baseline higher
BANKSY λ=0.2	+0.080	(−0.002, +0.176)	(−0.028, +0.227)	+0.05	+0.20	+0.28	−0.04	−0.04	unresolved
GraphST	+0.074	(−0.009, +0.170)	(−0.042, +0.223)	+0.03	+0.25	+0.24	−0.03	−0.04	unresolved
STAGATE	+0.054	(−0.000, +0.117)	(−0.018, +0.146)	+0.03	+0.15	+0.18	−0.01	−0.02	unresolved
HiSTaR	+0.052	(+0.004, +0.118)	(−0.007, +0.154)	+0.02	+0.06	+0.23	+0.01	−0.00	baseline higher
NichePCA	+0.037	(−0.028, +0.105)	(−0.061, +0.135)	+0.06	+0.13	+0.09	−0.07	−0.08	unresolved
AnisoST	+0.036	(−0.012, +0.088)	(−0.029, +0.114)	+0.01	+0.11	+0.13	+0.00	−0.04	unresolved
BANKSY λ=0.8	+0.033	(−0.013, +0.077)	(−0.029, +0.097)	+0.05	+0.04	+0.09	−0.06	+0.01	unresolved
BASS	+0.005	(−0.056, +0.077)	(−0.079, +0.120)	+0.08	+0.02	−0.06	−0.08	−0.07	unresolved

Bootstrap rank	P(rank 1)	median	95% interval
BASS	0.43	2	1-−10
Linear diffusion	0.45	2	1-−7
BANKSY λ=0.8	0.02	4	2-−9
NichePCA	0.04	4	1-−9
AnisoST	0.04	4	1-−7
HiSTaR	0.00	6	2-−10
STAGATE	0.00	6	3-−9
GraphST	0.01	8	3-−10
BANKSY λ=0.2	0.01	8	3-−10
SEDR	0.01	11	4-−16
SpaceFlow	0.00	12	7-−16
CCST	0.00	12	10-−16
DeepST	0.00	12	10-−15
SpaGCN	0.00	14	10-−16
PCA	0.00	14	11-−16
Spatial-MGCN	0.00	15	11-−16

Supplementary Table S8. Rank of each method in the hierarchical bootstrap with matched clustering (first three columns) and with default clustering (last three), on the sections completed by every method (BayesSpace, run on Visium only, is not ranked). The last rows give the ranking restricted to methods with a result on every section, which excludes the methods stopped at the memory cap on the largest Stereo-seq section.

Method	P(rank 1)	Median rank	95% interval	P(rank 1), default	Median rank, default	95% interval, default
BASS	0.50	2	1-–10	0.59	1	1-–7
Linear diffusion	0.29	2	1-–8	0.24	2	1-–5
BANKSY λ=0.8	0.06	3	1-–9	0.10	3	1-–8
NichePCA	0.06	4	1-–9	0.00	10	4-–15
AnisoST	0.04	4	1-–7	0.03	4	1-–6
STAGATE	0.01	6	2-–8	0.01	7	3-–12
HiSTaR	0.03	7	1-–10	0.00	8	4-–12
GraphST	0.01	8	3-–10	0.00	9	4-–13
BANKSY λ=0.2	0.01	8	3-–10	0.02	7	2-–11
SEDR	0.01	11	4-–16	0.00	6	3-–14
DeepST	0.00	11	10-–13	0.00	11	7-–13
SpaceFlow	0.00	12	7-–16	0.00	13	6-–16
SpaGCN	0.00	13	9-–15	0.00	14	7-–16
CCST	0.00	14	11-–16	0.00	11	8-–15
PCA	0.00	14	11-–16	0.00	15	11-–16
Spatial-MGCN	0.00	16	12-–16	0.00	14	11-–16
Methods with every section						
Linear diffusion	0.74	1	1-–5			
BANKSY λ=0.8	0.11	2	1-–6			
AnisoST	0.08	3	1-–5			
STAGATE	0.01	4	2-–6			
HiSTaR	0.04	5	1-–7			
BANKSY λ=0.2	0.02	6	2-–7			
SEDR	0.01	8	4-–11			
SpaceFlow	0.00	8	5-–11			
SpaGCN	0.00	9	6-–11			
CCST	0.00	10	8-–11			
PCA	0.00	10	8-–11			

Supplementary Table S9. Seed-to-seed standard deviation of ARI (mean over sections; default clustering). Methods run with one seed have no value.

Method	DLPFC	STARmap	MERFISH	Breast	Stereo-seq	Seeds per dataset
Linear diffusion	0.039	0.012	0.054	0.024	0.020	3/3/3/3/3
AnisoST	0.022	0.042	0.015	0.042	0.028	3/3/3/3/3
PCA	0.015	0.019	0.015	0.018	0.027	3/3/3/3/3
HiSTaR	0.047	0.180	0.053	0.049	0.028	3/3/3/3/3
GraphST	–	0.086	0.031	0.034	0.007	1/3/3/3/3
BANKSY λ=0.8	0.067	0.006	0.041	0.022	0.016	3/3/3/3/3
BANKSY λ=0.2	0.056	0.063	0.022	0.023	0.015	3/3/3/3/3
SpaGCN	0.038	0.030	0.016	0.020	0.025	3/3/3/3/3
STAGATE	–	–	–	0.016	0.021	1/1/1/3/3
SEDR	0.042	0.083	0.060	0.014	0.020	3/3/3/3/3
SpaceFlow	0.032	0.101	0.030	0.049	0.013	3/3/3/3/3
NichePCA	0.023	0.034	0.029	0.000	0.008	3/3/3/3/3
CCST	–	–	–	0.020	0.024	1/1/1/3/3
Spatial-MGCN	–	–	–	0.046	0.005	1/1/1/3/3
DeepST	0.042	0.030	0.028	–	–	3/3/3/1/1
BASS	–	–	–	0.010	0.017	1/1/1/3/3
BayesSpace	–	–	–	–	–	1/0/0/1/0

Supplementary Table S10. Initialisation experiment: one fixed embedding per section refitted 20 times with a single-start Gaussian mixture (up to 24 sections per method). Range and SD of ARI across the 20 fits; best-by-ARI minus mean: ARI gain of choosing the best fit by agreement with the labels; best-likelihood minus mean: gain of choosing the fit with the highest likelihood; ρ: median Spearman correlation between log-likelihood and ARI.

Method	Median range	IQR	Mean SD	Best-by-ARI minus mean	Best-likelihood minus mean	ρ DLPFC	ρ STARmap	ρ MERFISH	ρ Breast	ρ Stereo-seq
SpaceFlow	0.07	0.05-−0.10	0.027	0.036	+0.005	−0.05	+0.60	+0.51	−0.18	+0.10
GraphST	0.10	0.03-−0.15	0.031	0.045	+0.012	+0.36	+0.23	−0.04	−0.09	−0.13
CCST	0.11	0.08-−0.28	0.053	0.071	−0.052	−0.04	+0.26	−0.17	−0.42	+0.09
Spatial-MGCN	0.11	0.07-−0.15	0.031	0.047	−0.014	+0.07	+0.12	+0.19	−0.34	+0.45
DeepST	0.12	0.07-−0.22	0.043	0.060	−0.013	+0.14	+0.38	−0.26	+0.19	+0.14
PCA	0.13	0.08-−0.19	0.037	0.055	+0.017	−0.16	+0.17	−0.25	+0.25	−0.12
STAGATE	0.16	0.10-−0.23	0.049	0.067	+0.031	+0.62	+0.57	−0.22	−0.26	−0.18
SEDR	0.16	0.10-−0.21	0.055	0.092	+0.024	+0.50	−0.64	−0.20	−0.51	−0.31
AnisoST	0.16	0.13-−0.25	0.055	0.086	+0.037	+0.48	+0.47	−0.36	+0.13	+0.14
HiSTaR	0.17	0.13-−0.22	0.063	0.107	+0.056	+0.22	+0.53	−0.57	−0.00	−0.32
Linear diffusion	0.18	0.12-−0.24	0.057	0.071	+0.052	+0.88	+0.26	−0.01	+0.07	+0.28
BANKSY λ=0.2	0.18	0.11-−0.29	0.058	0.089	+0.033	+0.25	+0.57	+0.10	−0.16	+0.25
BANKSY λ=0.8	0.19	0.14-−0.28	0.071	0.091	+0.005	+0.19	+0.76	−0.08	+0.27	+0.25
NichePCA	0.19	0.13-−0.27	0.063	0.088	+0.026	+0.65	+0.54	+0.22	−0.26	−0.22

Supplementary Table S11. R mclust 6.0.1 (EEE) on fixed embeddings. Seed range: ARI range over ten R random seeds with the default initialisation, which uses a random subset of 2000 observations when there are more (sections with more than 2000 spots or cells only; STARmap sections are smaller and do not vary). Noise range: ARI range over five repetitions with 1% Gaussian noise when EM is initialised on all observations, which is deterministic. Full minus seeded mean: ARI of the full-data initialisation minus the mean of the ten seeded fits.

Method	Embeddings ($n > 2000$)	Seed range (mean)	Seed range (max)	Range >0.1	Noise range, full init.	Full minus seeded mean
Linear diffusion	20 of 23	0.17	0.35	75%	0.12	-0.000
NichePCA	20 of 23	0.17	0.42	75%	0.17	+0.007
PCA	20 of 23	0.15	0.46	55%	0.11	-0.018
SEDR	20 of 23	0.13	0.33	65%	0.08	+0.008
BANKSY λ=0.8	20 of 23	0.13	0.34	55%	0.10	-0.005
STAGATE	20 of 23	0.12	0.27	55%	0.08	+0.022
HiSTaR	20 of 23	0.11	0.32	55%	0.08	-0.005
AnisoST	20 of 23	0.10	0.22	45%	0.06	-0.029
Spatial-MGCN	19 of 22	0.10	0.40	37%	0.06	+0.010
BANKSY λ=0.2	21 of 24	0.09	0.28	48%	0.07	-0.003
CCST	20 of 23	0.09	0.35	20%	0.06	+0.011
SpaceFlow	20 of 23	0.09	0.20	40%	0.08	+0.001
GraphST	20 of 23	0.08	0.26	35%	0.06	-0.004
DeepST	20 of 23	0.05	0.17	20%	0.03	+0.008

Supplementary Table S12. Change in mean ARI when each method's default clustering is replaced by matched clustering (20-start best-likelihood Gaussian mixture) on the same embedding (all sections with both results). The two protocols can differ in algorithm and number of starts.

Method	Mean change in ARI	Sections improved	Wilcoxon p (unadjusted)
NichePCA	+0.077	16/23	0.023
HiSTaR	+0.042	17/24	0.006
STAGATE	+0.031	13/24	0.303
GraphST	+0.023	15/23	0.111
AnisoST	+0.014	14/24	0.121
Linear diffusion	+0.010	12/24	0.158
PCA	+0.010	9/24	0.406
BANKSY λ=0.2	+0.007	13/24	0.240
BANKSY λ=0.8	-0.001	9/24	0.987
SpaceFlow	-0.012	9/24	0.439
DeepST	-0.018	7/23	0.345
Spatial-MGCN	-0.033	8/23	0.035
CCST	-0.059	6/24	0.010
SEDR	-0.077	7/24	0.074

Supplementary Table S13. Metric panel from the standardised runner (shared preprocessing for all methods; mean over the five datasets of the per-dataset means). Values differ slightly from Supplementary Table S3 because the preprocessing is shared. Missed domains: annotated domains with no matched cluster; small-domain recall: recall of domains below 5% of spots; CHAOS and PAS: spatial fragmentation (lower is better). Runs: last column (successful runs; up to 72 = 24 sections $\times$ 3 seeds).

Method	ARI	NMI	Macro-F1	Missed domains	Small-domain recall	CHAOS↓	PAS↓	Boundary F1	Runs
BASS	0.549	0.651	0.608	2.21	0.637	1.064	0.038	0.668	69
Linear diffusion	0.530	0.653	0.609	2.15	0.670	1.032	0.009	0.627	72
BANKSY λ=0.8	0.518	0.629	0.591	2.14	0.650	1.041	0.027	0.628	72
BayesSpace	0.500	0.638	0.507	2.50	0.565	1.045	0.054	0.655	13
AnisoST	0.486	0.597	0.557	2.53	0.592	1.044	0.044	0.627	72
NichePCA	0.466	0.601	0.562	2.17	0.504	1.037	0.011	0.561	69
STAGATE	0.445	0.582	0.580	2.01	0.702	1.098	0.064	0.620	72
SEDR	0.442	0.596	0.553	2.24	0.533	1.056	0.040	0.630	72
BANKSY λ=0.2	0.432	0.540	0.523	2.63	0.581	1.238	0.161	0.605	72
SpaceFlow	0.423	0.561	0.519	2.39	0.440	1.055	0.029	0.580	72
HiSTaR	0.419	0.566	0.558	2.25	0.614	1.127	0.085	0.616	72
GraphST	0.399	0.536	0.537	2.05	0.413	1.114	0.108	0.621	69
CCST	0.394	0.510	0.453	3.39	0.408	1.080	0.044	0.554	71
DeepST	0.379	0.545	0.551	1.63	0.695	1.368	0.187	0.577	23
SpaGCN	0.352	0.463	0.467	2.48	0.559	1.292	0.256	0.615	72
PCA	0.350	0.450	0.462	2.85	0.520	1.352	0.275	0.608	72
Spatial-MGCN	0.348	0.469	0.467	2.38	0.617	1.319	0.217	0.571	69

Supplementary Table S14. Run time, peak memory and estimated energy of one whole job (data loading, embedding and clustering) in the standardised runner on one CPU thread. Energy is estimated at 10 W per busy core.

Method	Median time per job (s)	Peak memory (GB)	Energy per job (Wh, estimate)
BANKSY λ=0.8	23	4.8	0.07
BANKSY λ=0.2	24	4.8	0.07
AnisoST	24	3.7	0.07
Linear diffusion	25	3.7	0.08
PCA	26	3.7	0.08
NichePCA	58	9.1	0.16
SpaGCN	61	8.5	0.27
SEDR	106	9.4	0.34
HiSTaR	129	3.7	0.42
BASS	744	10.1	2.08
DeepST	791	10.8	2.56
SpaceFlow	949	3.9	2.86
STAGATE	1045	3.4	3.36
GraphST	1254	5.0	3.79
Spatial-MGCN	1515	7.9	4.49
BayesSpace	1892	12.5	5.49
CCST	2789	5.5	10.74

Supplementary Table S15. Mean ARI with each method's default clustering in its published pipeline (used in the main text for DLPFC, STARmap and MERFISH) and in the standardised runner (shared preprocessing, used for Supplementary Tables S13 and S14), and their difference (runner minus pipeline). For the three training-free baselines the runner's own clustering is the 20-start Gaussian mixture, so they are compared with the pipeline's matched-clustering value.

Method	DLPFC pipeline	DLPFC runner	DLPFC diff.	STARmap pipeline	STARmap runner	STARmap diff.	MERFISH pipeline	MERFISH runner	MERFISH diff.
Linear diffusion	0.558	0.569	+0.011	0.705	0.733	+0.028	0.497	0.497	+0.000
AnisoST	0.551	0.554	+0.004	0.622	0.621	−0.001	0.363	0.363	+0.000
PCA	0.409	0.417	+0.007	0.294	0.295	+0.001	0.130	0.130	+0.000
HiSTaR	0.491	0.497	+0.006	0.593	0.515	−0.077	0.221	0.237	+0.016
GraphST	0.514	0.502	−0.013	0.388	0.388	+0.000	0.251	0.260	+0.009
BANKSY λ=0.8	0.494	0.494	+0.000	0.740	0.740	+0.000	0.428	0.428	+0.000
BANKSY λ=0.2	0.514	0.514	−0.000	0.502	0.502	+0.000	0.211	0.211	+0.000
SpaGCN	0.422	0.422	+0.000	0.267	0.259	−0.008	0.151	0.113	−0.039
STAGATE	0.497	0.509	+0.011	0.479	0.550	+0.071	0.368	0.331	−0.037
SEDR	0.515	0.515	+0.000	0.593	0.593	+0.000	0.406	0.406	+0.000
SpaceFlow	0.312	0.312	+0.000	0.628	0.628	+0.000	0.389	0.389	+0.000
NichePCA	0.321	0.321	+0.000	0.692	0.692	+0.000	0.475	0.475	+0.000
CCST	0.437	0.438	+0.001	0.505	0.498	−0.007	0.212	0.222	+0.010
Spatial-MGCN	0.418	0.414	−0.004	0.303	0.304	+0.001	0.182	0.175	−0.006
DeepST	0.459	0.456	−0.003	0.368	0.343	−0.025	0.231	0.220	−0.011
BASS	0.484	0.490	+0.005	0.716	0.718	+0.001	0.582	0.561	−0.022
BayesSpace	0.437	0.437	+0.000	–	–	–	–	–	–

Supplementary Table S16. Sensitivity to the number of domains: mean ARI over all sections when the seed-0 embedding is clustered with matched clustering (20-start tied Gaussian mixture) for K from two below to two above the annotated number, and for the K in that range with the lowest BIC (label-free choice; it was almost always the largest K allowed).

Method	$K-2$	$K-1$	True K	$K+1$	$K+2$	BIC choice
Linear diffusion	0.466	0.536	0.542	0.494	0.462	0.462
AnisoST	0.417	0.490	0.506	0.475	0.432	0.432
BANKSY λ=0.8	0.404	0.444	0.487	0.512	0.487	0.487
NichePCA	0.439	0.479	0.481	0.497	0.472	0.472
STAGATE	0.417	0.466	0.480	0.463	0.425	0.433
HiSTaR	0.354	0.438	0.470	0.455	0.415	0.415
BANKSY λ=0.2	0.281	0.402	0.455	0.447	0.431	0.431
GraphST	0.378	0.451	0.447	0.406	0.397	0.395
SEDR	0.316	0.358	0.403	0.386	0.374	0.374
DeepST	0.279	0.358	0.366	0.390	0.386	0.385
SpaceFlow	0.338	0.384	0.365	0.341	0.339	0.339
PCA	0.283	0.301	0.343	0.330	0.325	0.327
CCST	0.215	0.276	0.323	0.359	0.370	0.371
Spatial-MGCN	0.284	0.308	0.311	0.322	0.317	0.320

Supplementary Table S17. Breast cancer scored against the 20 annotated regions (as in the main analysis) and against the 4 coarse classes they belong to (healthy, tumour, invasive, surrounding tumour). Seed-0 embedding, matched clustering (20-start tied Gaussian mixture) with K set to the number of regions or classes. Methods that cluster internally are not included.

Method	20 regions, $K = 20$	4 classes, $K = 4$
NichePCA	0.614	0.308
SpaceFlow	0.526	0.300
AnisoST	0.501	0.210
BANKSY λ=0.2	0.564	0.205
STAGATE	0.536	0.202
GraphST	0.607	0.197
SEDR	0.511	0.196
PCA	0.576	0.194
HiSTaR	0.494	0.190
CCST	0.533	0.187
Linear diffusion	0.529	0.186
BANKSY λ=0.8	0.604	0.181
DeepST	0.552	0.091
Spatial-MGCN	0.518	0.041

Supplementary Table S18. Reproduction of HiSTaR on the twelve DLPFC sections.

Setting	Median ARI	Mean ARI	95% CI of mean	Source
Reported (HiSTaR article)	0.65 (median)	–	–	HiSTaR article
Official code, 3-start GMM (3 seeds)	0.509	0.491	0.454-–0.525	this study
Masking at inference removed	0.502	0.487	0.441-–0.529	this study
R mclust as in the official pipeline (seed 0)	–	0.449	–	this study

Supplementary Table S19. Mean ARI of the final training epoch (used in the benchmark) and of the epoch with the highest ARI against the annotation, as selected by the released test scripts (seed 0).

Dataset	Sections	Final epoch	Label-selected epoch	Difference
DLPFC	12	0.418	0.551	+0.133
STARmap	3	0.303	0.687	+0.384
MERFISH	5	0.182	0.221	+0.039
All	20	0.342	0.489	+0.147

Supplementary Table S20. Automated screening of released code for patterns of label use (number of candidate lines per category; L1 result chosen by agreement with labels, L2 number of domains from labels, L3 unannotated spots removed, L4 settings keyed by section, L5 loops combined with a label metric). Candidate lines are not findings: each was read manually; the 14 repositories are the benchmarked methods with released code plus SPARK, SpatialPCA and conST, which were screened but not benchmarked, and only the label-selected epoch in Spatial-MGCN's test scripts was confirmed as L1 (Supplementary Note 5).

Repository	L1	L2	L3	L4	L5
BASS	0	0	0	0	0
Banksy_py	2	0	4	0	3
BayesSpace	0	0	2	0	0
CCST	0	5	0	0	1
DeepST	0	1	3	3	1
GraphST	0	0	0	0	0
SEDR	0	0	2	1	0
SPARK	0	0	7	0	0
STAGATE_pyG	0	0	0	0	0
SpaGCN	0	2	0	0	0
SpaceFlow	0	0	0	0	0
Spatial-MGCN	10	3	5	0	2
SpatialPCA	0	0	2	0	0
conST	0	0	2	0	0

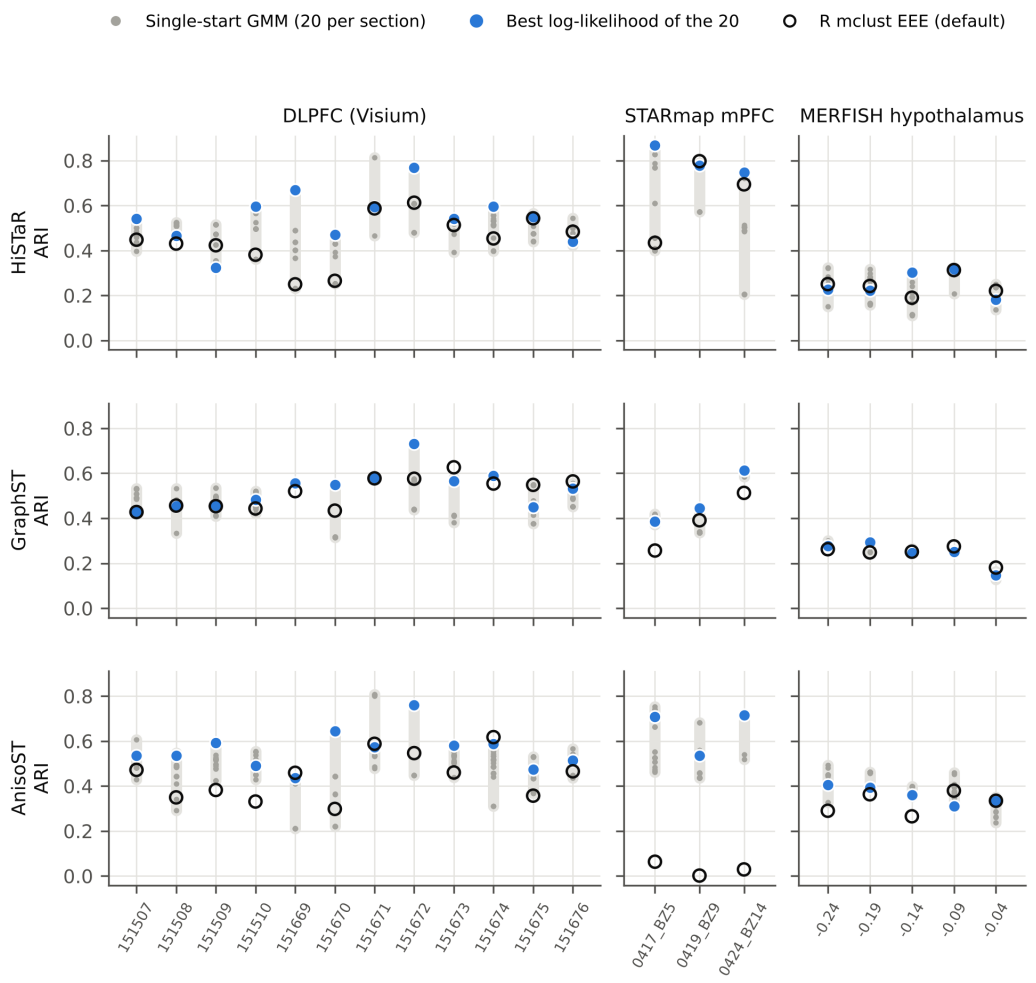
Supplementary Table S21. Pilot invariance test (three training-free methods, two sections): agreement (ARI) between the partition on the original data and on transformed data that should not change the answer. The methods are invariant to rescaling, rotating and reflecting the coordinates and to rescaling expression, but not fully to spot order, because the Gaussian mixture initialisation depends on it.

Section and method	permute spots	rescale coordinates x100	rescale coordinates x0.01	translate coordinates	rotate coordinates 37°	reflect coordinates	permute features	scale expression x2
DLPFC 151673 PCA	0.99	1.00	1.00	1.00	1.00	1.00	1.00	1.00
DLPFC 151673 Linear diffusion	0.97	1.00	1.00	1.00	1.00	1.00	1.00	1.00
DLPFC 151673 AnisoST	0.89	1.00	1.00	1.00	1.00	1.00	1.00	1.00
MERFISH −0.14 PCA	0.39	1.00	1.00	1.00	1.00	1.00	1.00	1.00
MERFISH −0.14 Linear diffusion	0.94	1.00	1.00	1.00	1.00	1.00	1.00	1.00
MERFISH −0.14 AnisoST	0.94	1.00	1.00	1.00	1.00	1.00	1.00	1.00

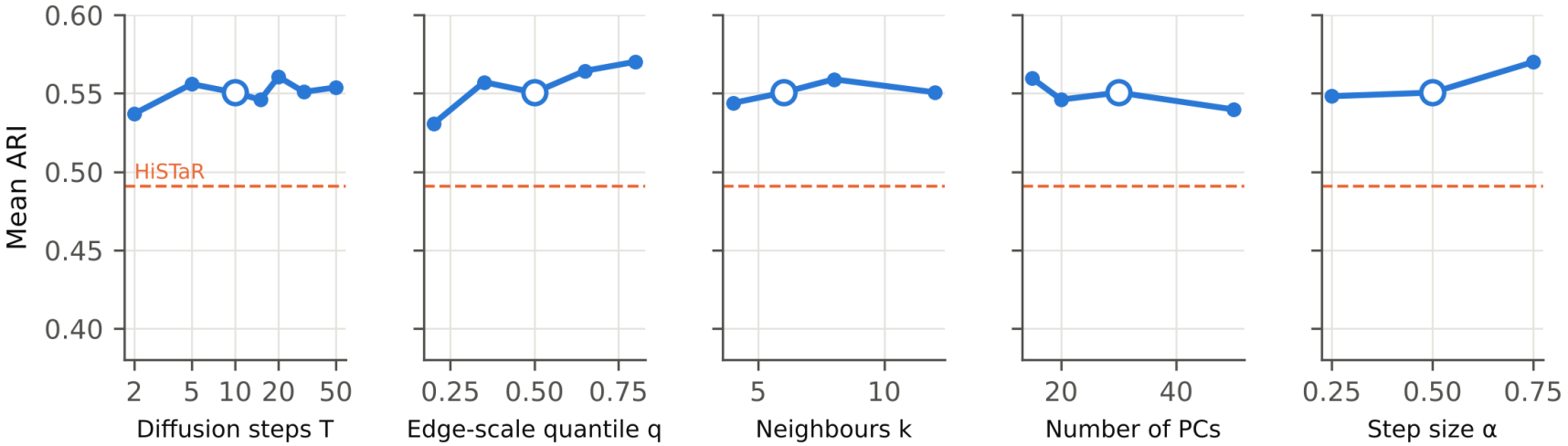
Supplementary Table S22. Which method was run in which experiment (sections completed / expected; seeds). n/a: the experiment needs an embedding and the method clusters internally.

Method	ARI, paper pipeline (sections / seeds)	Standardised runner + metric panel + resources (sections / seeds)	Lottery (20 starts)	R mclust perturbation	R mclust seeds	Invariance tests (sections)
Linear diffusion	20/24 / 3	24/24 / 3	24/24	20/24	23/24	2/24
AnisoST	20/24 / 3	24/24 / 3	24/24	20/24	23/24	2/24
PCA	20/24 / 3	24/24 / 3	24/24	20/24	23/24	2/24
BANKSY λ=0.2	20/24 / 3	24/24 / 3	24/24	20/24	24/24	not run
BANKSY λ=0.8	20/24 / 3	24/24 / 3	24/24	20/24	23/24	not run
HiSTaR	20/24 / 3	24/24 / 3	24/24	20/24	23/24	not run
GraphST	20/24 / 3	23/24 / 3	23/24	20/24	23/24	not run
SpaGCN	20/24 / 3	24/24 / 3	n/a (no embedding)	n/a (no embedding)	n/a (no embedding)	not run
STAGATE	20/24 / 1	24/24 / 3	24/24	20/24	23/24	not run
SEDR	20/24 / 3	24/24 / 3	24/24	20/24	23/24	not run
SpaceFlow	20/24 / 3	24/24 / 3	24/24	20/24	23/24	not run
NichePCA	20/24 / 3	23/24 / 3	23/24	20/24	23/24	not run
CCST	20/24 / 1	24/24 / 3	24/24	20/24	23/24	not run
Spatial-MGCN	20/24 / 1	23/24 / 3	23/24	20/24	22/24	not run
DeepST	20/24 / 3	23/24 / 1	23/24	20/24	23/24	not run
BASS	20/24 / 1	23/24 / 3	n/a (no embedding)	n/a (no embedding)	n/a (no embedding)	not run
BayesSpace	12/13 / 1	13/13 / 1	n/a (no embedding)	n/a (no embedding)	n/a (no embedding)	not run

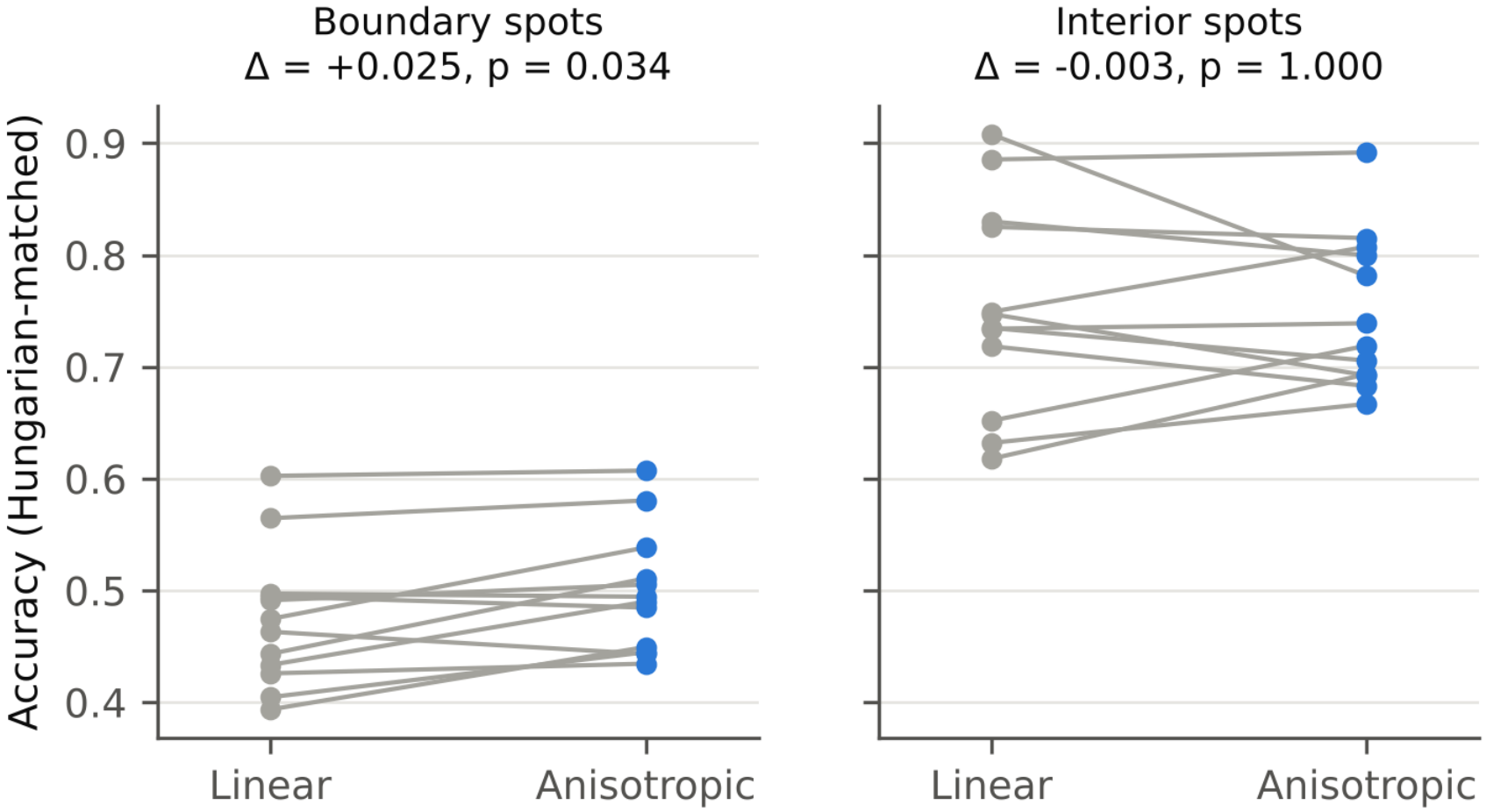
Supplementary Figures



Supplementary Figure S1. Initialisation experiment for three representative methods (HiSTaR, GraphST and AnisoST), per section: ARI of 20 single-start fits of one fixed embedding (grey), the best-likelihood fit (blue) and R mclust with default settings (open circle).



Supplementary Figure S2. Sensitivity of linear diffusion (diffusion steps T, neighbours k, number of PCs, step size α) and of AnisoST (edge-scale quantile q) on the 12 DLPFC sections, one hyper-parameter at a time (open circle: default; dashed line: HiSTaR mean ARI).



Supplementary Figure S3. Accuracy on spots at layer boundaries and in layer interiors on the 12 DLPFC sections, linear diffusion versus AnisoST (labelled Anisotropic); one line per section; Wilcoxon signed-rank test, $n = 12$ sections.